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Loader latching assist method and apparatus

Abstract

A method and apparatus assists coupling a loader with a work vehicle. A pre-position instruction signal is generated to pre-position the loader in a predetermined orientation to facilitate the coupling. A forward movement instruction signal is generated for initiating movement of the associated work vehicle **1** towards the loader. A coupled confirmation signal representative of the loader being coupled with the associated work vehicle is received. A forward movement pause instruction signal for pausing the movement of the associated work vehicle towards the loader is generated responsive to receiving the coupled confirmation signal. One or more of the pre-position instruction signal, the forward movement instruction signal, and/or the forward movement pause instruction signal may be automated signals delivered to the work vehicle for executing the commands or images rendered on a human readable display unit for instructing an operator actions to take for coupling the loader with the work vehicle.

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Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure relates to loaders and work vehicles and, in particular, to methods and apparatus assisting coupling loaders with work vehicles such as tractors.

BACKGROUND

(2) Loader arrangements, i.e., the arrangement of a loader on a work vehicle, in particular on an agricultural vehicle, or on different types of utility vehicles, are known. Loaders can be connected to, or mounted on, a vehicle, for example onto a tractor or an agricultural or other vehicle by means of a mounting frame, in order to carry out loader operations. The mounting frame is customarily bolted or otherwise secured or connected to the vehicle frame or fastened to the latter for selective removal as necessary or desired. The loader itself has a corresponding connecting or mounting point or mounting mast that may be selectively connected to, or mounted on, or coupled to, the mounting frame on the work vehicle.

(3) For efficient work flow, operators prefer to drive the work vehicle to a position adjacent to the loader, then mechanically engage one or more connecting or mounting points of the loader with a corresponding one or more connecting or mounting points of the mounting frame on the work vehicle, all without leaving the operator's station of the work vehicle or, at a minimum, all by leaving the operator's station of the tractor only once such as to couple the vehicle hydraulics with corresponding hydraulic subsystems of the loader. Operators also prefer that they are able to establish the desired mutual mechanical connection between the loader and the mounting frame on the work vehicle on their first attempt. This is commonly referred to as a "1-trip" connection. True 1-trip connections are seldom realized in practice for several reasons including misalignment between the connecting or mounting points of the loader and mounting frame. Geometric alignment between connection points of the vehicle and those of the loader arrangement can sometimes change during a time in which the loader arrangement was decoupled from the vehicle for reasons including location instabilities such as for example ground unevenness, subsidence of the ground, etc.

(4) Typical loaders have two parallel arms that extend at the front end of the tractor and that are equipped with a corresponding loading tool, such as a shovel, a gripper, a loading fork, etc. The arms and the loader tool are usually operable by hydraulic or electrical actuators. Apparatus such as parking stands have been proposed that help to stage or otherwise pre-position such loader

arrangements prior to attempting to couple the loader onto the tractor. Parking stand units are available that typically comprise pivotable parking support members that are provided on the underside of the arms of the front loader. The pivotable parking support members can be brought into a lowered parking position, and locked in the parking position wherein, in the parking position, the front loader can be decoupled from the tractor and can thereby be supported on the parking support in such a manner that the arms are held in an upright coupling and decoupling position when the front loader is so parked. In that way, decoupling or coupling the front loader is made possible by releasing (or closing) a front loader lock and subsequent (or previous) maneuvering of the tractor.

(5) While parking stands have been helpful in improving the efficiency of loader connection work flows, they still fall short of providing consistent 1-trip connection operation because of persistent misalignment between the mounting frame on the work vehicle and the connecting or mounting point of the loader.

(6) Another approach for improving loader attachment efficiency is focused on the interface between the loader and the tractor. To this end, arrangements have been proposed for effecting a float condition in loader hydraulic lift cylinders after the hydraulic fluid supply to the loader is decoupled. The mounting frame on the work vehicle can, for example, comprise a ramp along which the loader mounting mast may be guided and displaced for the coupling and decoupling. By means of the floating condition, the mounting mast can be pivoted about a coupling point thereof with respect to the loader arm, i.e., relative to the loader arm, and, for example during the coupling, can be adapted to a positioning angle with respect to the mounting frame or aligned with respect to the latter. An alignment takes place by bearing bolts of the mounting mast sliding over the ramp during the coupling and being guided into the catch hooks, wherein, by means of this interaction, the bearing bolts are brought into the position thereof provided for the coupling. Without the above-mentioned floating position of the hydraulic cylinder, the mounting mast would be in a rigid connection with respect to the loader arm such that the mounting mast cannot carry out any pivoting movements relative to the loader arm making mutual engagement difficult.

(7) While effecting the float condition in hydraulic lift cylinders of loaders have been helpful in improving the efficiency of work flows, they also may still fall short of providing consistent 1-trip connection operation for reasons including because of the persistent misalignment mentioned above between the connecting or mounting points of the mounting frame on the work vehicle and the connecting or mounting points of the loader.

SUMMARY

(8) In accordance with an aspect, a method is provided for assisting coupling a loader with an associated work vehicle. In any of the embodiments herein, portions of the method for assisting coupling the loader with the associated work vehicle may be performed with the loader and the associated work vehicle disposed in a mutually spaced apart arrangement, and other portions may be performed with the loader and the associated work vehicle disposed in a mutual contact arrangement. The loader and the associated work vehicle may be disposed in the mutually spaced apart arrangement when the loader is resting or otherwise supported by parking stands or the like, for example. In the method and with the loader and the associated work vehicle mutually spaced apart, a pre-position instruction signal is generated to pre-position the loader in a predetermined orientation to facilitate the coupling. In the method and also with the loader and the associated work vehicle mutually spaced apart, a forward movement instruction signal is generated for initiating movement of the associated work vehicle towards the loader. Then, with the loader and the associated work vehicle being in mutual contact, a coupled confirmation signal is received, wherein the coupled confirmation signal is representative of a confirmation of the loader being coupled with the associated work vehicle. The loader and the associated work vehicle being in mutual contact may occur, for example, following or subsequent to the movement of the associated work vehicle towards the loader. In the method and also with the loader and the associated work

vehicle being in mutual contact, a forward movement pause instruction signal is generated responsive to receiving the coupled confirmation signal, wherein the forward movement pause instruction signal is generated for pausing the movement of the associated work vehicle towards the loader.

(9) In any of the embodiments herein, the generating the pre-position instruction signal includes generating a pre-position instruction image on a screen of a human readable display unit, wherein the pre-position instruction image informs an operator of the associated vehicle how to pre-position the loader in the predetermined orientation to facilitate the coupling. In addition and in any of the embodiments herein, the generating the forward movement instruction signal includes generating a forward movement instruction image on the screen of a human readable display unit, wherein the forward movement instruction image instructs the operator of the associated work vehicle to initiate the movement of the associated work vehicle towards the loader. In addition and in any of the embodiments herein, the receiving the coupled confirmation signal includes receiving a manual coupled confirmation signal from the operator of the associated vehicle via an input device confirming by the operator that the loader is coupled with the associated work vehicle. In addition and in any of the embodiments herein, the generating the forward movement pause instruction signal includes generating, responsive to receiving the coupled confirmation signal, a forward movement pause instruction image on the screen of the human readable display unit, wherein the forward movement pause instruction image instructs the operator of the associated work vehicle to pause the movement of the associated work vehicle towards the loader.

(10) In any of the embodiments herein, the method for assisting coupling a loader with an associated work vehicle further includes displaying an orientation image on the screen of the human readable display unit, wherein the orientation image is representative of desired orientations of one or more components of the loader informing the operator of the associated vehicle how to move the one or more components of the loader to predetermined positions for pre-positioning the loader in the predetermined orientation to facilitate the coupling.

(11) In any of the embodiments herein, the displaying the orientation image includes displaying an orientation image on the screen of the human readable display unit of one or more of an orientation of a carrier arm of the loader, an orientation of a mast arm of the loader, and/or an orientation of a mounting mast of the loader that is operably coupleable with a corresponding mounting frame affixed to the associated work vehicle.

(12) In any of the embodiments herein, the receiving the coupled confirmation signal includes receiving the coupled confirmation signal as a loader locked signal from a loader locked sensor disposed at an interface between a mounting mast of the loader that is operably coupleable with a corresponding mounting frame affixed to the associated work vehicle, wherein the loader locked signal received from the loader locked sensor is representative of a confirmation of the loader being coupled with the associated work vehicle.

(13) In any of the embodiments herein, the method for assisting coupling a loader with an associated work vehicle further includes generating a float condition instruction signal with the loader and the associated work vehicle being in mutual contact and with the associated work vehicle moving towards the loader, wherein the generating the float condition instruction signal is to establish a float condition in hydraulic lifting cylinders of the loader to relax a mounting mast of the loader that is operably coupleable with a corresponding mounting frame affixed to the associated work vehicle permitting the mounting mast to move relative to the mounting frame during the coupling.

(14) In any of the embodiments herein, the generating the float condition instruction signal includes generating a float condition cycle instruction image on the screen of the human readable display unit, wherein the float condition cycle instruction image instructs the operator of the associated work vehicle to intermittently establish, with the loader and the associated work vehicle being in mutual contact and with the associated work vehicle moving towards the loader, the float condition

between periods of a locked condition of the mounting mast of the loader wherein in the locked condition movement between the mounting mast and the mounting frame is prevented.

(15) In any of the embodiments herein, the generating the float condition instruction signal includes delivering the float condition cycle instruction signal via a communication interface to a controller the associated work vehicle to automatically intermittently establish, with the loader and the associated work vehicle being in mutual contact and with the associated work vehicle moving towards the loader, the float condition between periods of a locked condition of the mounting mast of the loader wherein in the locked condition movement between the mounting mast and the mounting frame is prevented.

(16) In any of the embodiments herein, the generating the pre-position instruction signal includes delivering the pre-position instruction signal via a communication interface to a controller the associated work vehicle to automatically pre-position the loader in the predetermined orientation to facilitate the coupling. In addition and in any of the embodiments herein, the generating the forward movement instruction signal includes delivering the pre-position instruction signal via the communication interface to the controller the associated work vehicle to automatically initiate the movement of the associated work vehicle towards the loader. In addition and in any of the embodiments herein, the receiving the coupled confirmation signal includes automatically receiving the coupled confirmation signal as a loader locked signal from a loader locked sensor disposed at an interface between a mounting mast of the loader that is operably coupleable with a corresponding mounting frame affixed to the associated work vehicle, wherein the loader locked signal received from the loader locked sensor is representative of a confirmation of the loader being coupled with the associated work vehicle. In addition and in any of the embodiments herein, the generating the forward movement pause instruction signal includes delivering, responsive to receiving the coupled confirmation signal, the forward movement pause instruction signal via the communication interface to the controller the associated work vehicle to automatically pause the movement of the associated work vehicle towards the loader.

(17) In accordance with an aspect, an attach assist apparatus is provided for assisting coupling a loader with an associated work vehicle. In an example embodiment, the apparatus includes an attach assist control unit including a processor device, a non-transient memory device operatively coupled with the processor device, and attach assist control logic stored in the memory device. In the example embodiment, the processor device is operable to execute the attach assist control logic to generate a pre-position instruction signal for pre-positioning the loader in a predetermined orientation to facilitate the coupling. The processor device is further operable to execute the attach assist control logic to generate a forward movement instruction signal for initiating movement of the associated work vehicle towards the loader. The processor device is still further operable to execute the attach assist control logic to receive a coupled confirmation signal representative of a confirmation of the loader being coupled with the associated work vehicle. The processor device is yet still further operable to execute the attach assist control logic to generate based on receiving the coupled confirmation signal a forward movement pause instruction signal for pausing the movement of the associated work vehicle **1** towards the loader.

(18) In any of the embodiments herein, the attach assist apparatus further includes a human readable display unit operably coupled with the attach assist control unit, and an input device operably coupled with the attach assist control unit. The human readable display unit includes a screen operable to display images that are viewable by an operator of the associated work vehicle.

(19) The processor device is operable to execute the attach assist control logic to generate the pre-position instruction signal by generating a pre-position instruction image and displaying the pre-position instruction image on the screen of the human readable display unit, wherein the pre-position instruction image informs an operator of the associated vehicle how to pre-position the loader in the predetermined orientation to facilitate the coupling. The processor device is operable to execute the attach assist control logic to generate the forward movement instruction signal by

generating a forward movement instruction image, and displaying the forward movement instruction image on the screen of the human readable display unit, wherein the forward movement instruction image instructs the operator of the associated work vehicle to initiate the movement of the associated work vehicle towards the loader. The processor device is operable to execute the attach assist control logic to receive the coupled confirmation signal by receiving a manual coupled confirmation signal by the input device from the operator of the associated vehicle, the coupled confirmation signal confirming by the operator that the loader is coupled with the associated work vehicle. The processor device is operable to execute the attach assist control logic to generate the forward movement pause instruction signal by generating, responsive to receiving the coupled confirmation signal, a forward movement pause instruction image and displaying the forward movement pause instruction image on the screen of the human readable display unit, wherein the forward movement pause instruction image instructs the operator of the associated work vehicle to pause the movement of the associated work vehicle towards the loader.

(20) In any of the embodiments herein, the processor device of the attach assist apparatus is operable to execute the attach assist control logic to display an orientation image on the screen of the human readable display unit, wherein the orientation image is representative of desired orientations of one or more components of the loader informing the operator of the associated vehicle how to move the one or more components of the loader to predetermined positions for pre-positioning the loader in the predetermined orientation to facilitate the coupling.

(21) In any of the embodiments herein, the processor device of the attach assist apparatus is operable to execute the attach assist control logic to display the orientation image on the screen of the human readable display unit as one or more of an orientation of a carrier arm of the loader, an orientation of a mast arm of the loader, and/or an orientation of a mounting mast of the loader that is operably coupleable with a corresponding mounting frame affixed to the associated work vehicle.

(22) In any of the embodiments herein, the attach assist apparatus further includes a sensor input unit operably coupled with the attach assist control unit, and a loader locked sensor operably coupled with the attach assist control unit by the sensor input unit, wherein the loader locked sensor is disposed at an interface between a mounting mast of the loader that is operably coupleable with a corresponding mounting frame **16** affixed to the associated work vehicle. The processor device of the attach assist apparatus is operable to execute the attach assist control logic to receive the coupled confirmation signal by receiving a loader locked signal from the loader locked sensor as the coupled confirmation signal representative of the confirmation of the loader being coupled with the associated work vehicle.

(23) In any of the embodiments herein, the processor device of the attach assist apparatus is operable to execute the attach assist control logic to generate a float condition instruction signal to establish a float condition in hydraulic lifting cylinders of the loader to relax a mounting mast of the loader that is operably coupleable with a corresponding mounting frame affixed to the associated work vehicle permitting the mounting mast to move relative to the mounting frame during the coupling.

(24) In any of the embodiments herein, the processor device of the attach assist apparatus is operable to execute the attach assist control logic to generate the float condition instruction signal by generating a float condition cycle instruction image on the screen of the human readable display unit, wherein the float condition cycle instruction image instructs the operator of the associated work vehicle to intermittently establish, with the loader and the associated work vehicle being in mutual contact and with the associated work vehicle moving towards the loader, the float condition between periods of a locked condition of the mounting mast of the loader wherein in the locked condition movement between the mounting mast and the mounting frame is prevented.

(25) In any of the embodiments herein, the attach assist apparatus further includes a control signal output unit operably coupled with the attach assist control unit, and a valve operably coupled with

the attach assist control unit by the control signal output unit, wherein the valve is responsive to a control signal generated by the attach assist apparatus to actuate to selectively relieve hydraulic fluid from a chamber of one or more hydraulic lifting cylinders of the loader. The processor device is operable to execute the attach assist control logic to generate the float condition instruction signal as the control signal to establish a float condition in hydraulic lifting cylinders of the loader to relax a mounting mast of the loader that is operably coupleable with a corresponding mounting frame affixed to the associated work vehicle permitting the mounting mast to move relative to the mounting frame during the coupling.

(26) In any of the embodiments herein, the attach assist apparatus further includes a communication interface operably coupling the attach assist control unit with a controller of the associated work vehicle, and a loader locked sensor operably coupled with the attach assist control unit by the sensor input unit, wherein the loader locked sensor is disposed at an interface between a mounting mast of the loader that is operably coupleable with a corresponding mounting frame affixed to the associated work vehicle. The processor device is operable to execute the attach assist control logic to generate the pre-position instruction signal by delivering the pre-position instruction signal via the communication interface to the controller of the associated work vehicle to automatically pre-position the loader in the predetermined orientation to facilitate the coupling.

(27) The processor device is operable to execute the attach assist control logic to generate the forward movement instruction signal by delivering the pre-position instruction signal via the communication interface to the controller of the associated work vehicle to automatically initiate the movement of the associated work vehicle towards the loader.

(28) The processor device is operable to execute the attach assist control logic to receive the coupled confirmation signal by automatically receiving the coupled confirmation signal as a loader locked signal from the loader locked sensor, the loader locked signal from the loader locked sensor being representative of the confirmation of the loader being coupled with the associated work vehicle.

(29) The processor device is operable to execute the attach assist control logic to generate the forward movement pause instruction signal by delivering, responsive to receiving the coupled confirmation signal, the forward movement pause instruction signal via the communication interface to the controller of the associated work vehicle to automatically pause the movement of the associated work vehicle towards the loader.

(30) Other embodiments, features and advantages of the example embodiments for assisting coupling loaders with work vehicles such as tractors will become apparent from the following description of the embodiments, taken together with the accompanying drawings, which illustrate, by way of example, the principles of the example embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the accompanying drawings which are incorporated in and constitute a part of the specification, embodiments of the invention are illustrated, which, together with a general description of the invention given above, and the detailed description given below, serve to exemplify the embodiments of this invention.

(2) FIG. 1 is a schematic side view of an agricultural vehicle with a front loader arrangement.

(3) FIG. 2 shows the front loader of FIG. 1 in a perspective schematic side view with its parking stand unit deployed with the front loader separated from the tractor and resting in a freestanding parked position.

(4) FIG. 3 is a schematic side view of a mounting mast in cross section and of a mounting frame of the front loader arrangement of FIGS. 1 and 2 in a decoupled state.

- (5) FIG. 4 is a further schematic side view of the mounting mast of the mounting frame of FIGS. 1 to 3 in a coupled state.
- (6) FIG. 5 is a schematic hydraulic circuit diagram for a front loader arrangement with single-acting hydraulic cylinders.
- (7) FIG. 6 is a schematic hydraulic diagram for a front loader arrangement with double-acting hydraulic cylinders.
- (8) FIG. 7 is a block diagram that illustrates a representative attach assist apparatus according to the example embodiments.
- (9) FIG. 8 is a flow diagram illustrating a method of assisting attaching a loader onto a work vehicle in accordance with an example embodiment.
- (10) FIGS. 9A-9F illustrations showing instructional images displayed on a human readable display device for assisting coupling a loader with a work vehicle such a tractor.

DETAILED DESCRIPTION

- (11) In the following description reference is made to the accompanying figures which form a part thereof, and in which is shown, by way of illustration, one or more example embodiments of the disclosed loader latching assist method and apparatus. Various modifications of the example embodiments may be contemplated by one of skill in the art.
- (12) As used herein, unless otherwise limited or modified, lists with elements that are separated by conjunctive terms (e.g., “and”) and that are also preceded by the phrase “one or more of” or “at least one of” indicate configurations or arrangements that potentially include individual elements of the list, or any combination thereof. For example, “at least one of A, B, and C” or “one or more of A, B, and C” or “one or more of A, B, and/or C” indicates the possibilities of only A, only B, only C, or any combination of two or more of A, B, and C (e.g., A and B; B and C; A and C; or A, B, and C).
- (13) Referring now to the drawings, wherein the showings are only for the purpose of illustrating the example embodiments only and not for purposes of limiting the same, FIG. 1 illustrates a work vehicle 1 in the form of an agricultural tractor 10 (carrier vehicle). The tractor 10 has a mounted loader arrangement 11 that may be a front loader arrangement 12 as shown, a wheeled loader arrangement, a telescopic loader arrangement, a rear loader arrangement or any other type loader arrangement. The loader arrangement 11 may have a boom as shown comprising a boom of a front loader or of wheeled loader, or as a jib of a telescopic loader, a rear loader, or the like. Front and rear loaders may be used as attachment units on agricultural tractors or construction machines. The expression “loader” herein is also intended to encompass other loader-like assemblies and machines that have a loader boom or a jib, for example construction machines such as diggers, bulldozers or forestry machines for deforestation of woodland, or if appropriate also cranes.
- (14) In the illustrated example, the front loader arrangement 12 comprises a front loader 13, mounting frames 16, and mounting masts 18. The mounting frames 16 are mounted on both sides of a vehicle frame 14 of the tractor 10, for the coupling of the front loader 13 with the tractor 10. The front loader 13 has the mounting masts 18, likewise on both sides of the tractor 10, by means of which the front loader 13 is coupled to the mounting frames 16 via the mounting masts 18, and therefore onto the tractor 10.
- (15) The front loader 13 has front loader arms 20 having a rear ends to the left as viewed in the Figure, that are pivotally fastened to the mounting masts 18 and forward ends to the right as viewed in the Figure to which a front loader tool 22 is pivotally connected. The front loader arms 20 extend on opposite sides of the tractor and comprise mast arms 19 and carrier arms 21 that run in parallel on the opposite lateral sides of the tractor 10. The mast arms 19 and carrier arms 21 may be connected to each other via one or more transverse carriers (not shown) as may be necessary or desired for providing a robust and rigid construction. The front loader tool 22 is designed, by way of example, in the form of a loading shovel or bucket 23, wherein the front loader tool 22 could also be designed as any type of work implement including loading forks, grippers, etc. The mast

arms **19** of the front loader arms **20** are pivotable via a pair of hydraulic lifting cylinders **24** that respectively extend at opposite lateral sides of the tractor **10** between the respective mounting masts **18** and the respective carrier arms **21** of the front loader arm **20**. A tilting linkage **26** serves for the parallel guidance of the front loader tool **22**, and extends in each case between links **25** that are pivotally connected to the carrier arms **21** of the front loader arms **20** provided on the opposite lateral sides of the tractor **10**, and the respective mounting masts **18**. The front loader tool **22** is pivotable on both sides of the front loader arms **20** via a pivoting linkage **27**, which is connected to the front loader arms **20** and to the front loader tool **22**, and also via a hydraulic pivoting cylinder **28** which is connected to the respective pivoting linkage **27** and to the respective link **25**.

(16) The mounting frames **16** of the front loader arrangement **12** each have a receiving region **17** for receiving, or for the coupling of or with, the mounting mast **18**. The mounting frames **16** of the front loader arrangement **12** further each have a fastening region **29** for fastening the mounting frames **16** to the vehicle frame **14**. At the fastening region **29**, the mounting frames **16** are screwed or otherwise bolted or fastened to the vehicle frame **14** via suitable connectors (not shown). The mounting frames **16** each have a first (front) catch hook **36** on a lower front region of the receiving region **17**, and a second (rear) catch hook **38** on an upper rear region of the receiving region **17**, or hook-shaped bearing points in each case.

(17) The tractor **10** and loader arrangement **11** may be used in combination with an attach assist apparatus **100** according to example embodiments of the present disclosure. The attach assist apparatus **100** of the example embodiments herein may execute or otherwise perform a loader attach assist method providing assistance to an operator for coupling the loader with the tractor and, in particular, for providing assistance to the operator for coupling the mounting mast **18** of the loader **13** onto the mounting frames **16** affixed to the tractor **10**, according to examples of the present disclosure.

(18) To provide a useful, albeit non-limiting example, the attach assist apparatus **100** of the example embodiments is described below in conjunction with a particular type of work vehicle **1**, shown in the form of an agricultural tractor **10**, equipped with a particular type of front loader tool **22** carrying an implement such as the bucket implement **23**. However, it is to be appreciated that the attach assist apparatus **100** of the example embodiments may be used in conjunction with any type of work vehicle equipped with any type of loader carrying any type of implement. In addition it is to be further appreciated that the attach assist apparatus **100** may be distributed as an integral part of a work vehicle **1**, as an integral part of a loader arrangement **11** or other module that may be removably attached to and detached from a work vehicle **1** on an as-needed basis, or as a discrete assembly or multi-component kit that may be installed on an existing work vehicle via retrofit attachment.

(19) It is to be appreciated that the motion of the hydraulic lifting cylinders **24** and the hydraulic pivoting cylinders **28** may be controlled directly by operator manipulation of various levers, pedals and/or other human interface device(s) operatively coupled with hydraulic control components **69** of the loader arrangement **11**. It is further to be appreciated that the motion of the loader arrangement **11** may be indirectly controlled by the operator rendering motion commands to via the attach assist apparatus **100** which in turn may operate the various hydraulic control components **69** of the loader arrangement **11** for moving the cylinders **24**, **28** and thereby effecting the desired movement of the loader arrangement **11**. The attach assist apparatus **100** may directly control the various hydraulic control components **69** of the loader arrangement **11** for moving the cylinders **24**, **28** and thereby effecting the desired movement of the loader arrangement **11**. The attach assist apparatus **100** also may indirectly control the various hydraulic control components **69** of the loader arrangement **11** by sending signals and/or commands to an electronic control module (ECM) **135** native to the associated work vehicle **1** via a communication link or the like for moving the cylinders **24**, **28** by the ECM **135** executing the signals and/or commands and thereby effecting the desired movement of the loader arrangement **11**. It is still further to be appreciated that the motion

of the hydraulic lifting cylinders **24** and/or hydraulic pivoting cylinders **28** may be controlled by the attach assist apparatus **100** automatically and/or semi-automatically in response to generalized operator commands such as for example return to position (RTP) commands wherein for example the attach assist apparatus **100** may function in an RTP mode to automatically return the mounting mast **18** of the front loader **13** to a position favorable to facilitate 1-step connection of the mounting mast **18** with the mounting frames **16** on the tractor **10**, thereby attaching the loader **13** with the tractor **10**. In addition, the mounting mast **18** of the front loader **13** may be returned to one or more pre-stored position(s) in response to operator input with the assistance of the attach assist apparatus **100** providing suitable instructions for helping to assist the operator in facilitating 1-step connection of the mounting mast **18** with the mounting frames **16** on the tractor **10**, thereby attaching the loader **13** with the tractor **10**.

(20) The attach assist apparatus **100** is only partially shown in FIG. **1** and an example embodiment of which will be described more fully below in conjunction with FIG. **7**. First, however, a general description of the use of parking stands in accordance with example embodiments for improving the efficiency of work flows providing 1-trip connection operation will be described. In this connection, FIG. **2** shows the front loader **13** of FIG. **1** in a perspective side view separated from the tractor, and resting in a freestanding parked position. As shown, the front loader **13** comprises a parking stand unit **40** arranged on the carrier arms **21** of the front loader arms **20**.

(21) The parking stand unit **40** of the example embodiment comprises a parking support **41** pivotably articulated on the carrier arms **21** and a locking brace **42**, likewise pivotably articulated at one end on the carrier arms **21** and also movably connected at the other end to the parking support **41**. The locking brace **42** can be brought into a locking position (as shown in FIG. **2**) on the parking support **41**, which position it assumes as soon as the parking support **41** has been pivoted into a parking position as illustrated. In the parking position or locking position, the parking support **41** is supported and locked relative to the front loader carrier arms **21** in a position determined by the length and the pivot angle of the locking brace **42**.

(22) At the free end of the parking support **41** an oscillating foot **43** is provided, which is mounted by means of a suitable fastener such as a bolt or the like so as to pivot back and forth on the parking support **41**. The foot **43** provides a standing or support surface when the parking stand unit **40** is pivoted into the parking or parked position as illustrated in FIG. **2**.

(23) Also provided with regard to the parking stand unit **40** of the example embodiment is a catch device **44** that is mounted in the form of a plate furnished with a catch notch **45** on the underside of the front loader carrier arms **21** in such a manner that a latch (not shown) engages in the catch notch **45** in the operating position due to its initial spring tension, thereby holding the parking stand unit **40** in the operating position or locks it in the operating position. By simply actuating the latch inside an elongated hole against an initial spring tension (moving in the longitudinal direction relative to the parking support), the parking stand unit **40** can be unlocked from being held in a stowed away position under the front loader carrier arms **21**, and brought into the parking position as illustrated in FIG. **2**.

(24) Even though the front loader **13** is detached from the tractor **10** such as shown in FIG. **2**, it is selectively animated in accordance with the example embodiments herein for reasons and in ways to be described in greater detail below by supplying hydraulic energy from a source **48** to one or more of the hydraulic lifting cylinders **24** and/or to the hydraulic pivoting cylinders **28**. That is, the loader **13** is selectively animated with the loader **13** and the associated work vehicle **1** mutually spaced apart and while resting in the freestanding parked position as shown in FIG. **2**. In the example embodiment, a multi-coupler device **50** is responsive to a control signal **150** from the attach assist apparatus **100** to direct a supply hydraulic fluid **51** from the source **48** to the hydraulic lifting cylinders **24** via one or more hydraulic lifting cylinder fluid coupling lines **24'** and/or to the hydraulic pivoting cylinders **28** via one or more hydraulic pivoting cylinder fluid coupling lines **28'**. The hydraulic lifting and pivoting cylinders **24**, **28** may be double acting cylinders but are

shown and described using only a single supply line for convenience and ease of illustration and description.

(25) The multi-coupler device **50** comprises first and second mutually coupleable portions **50a**, **50b** in the example embodiment, wherein the first and second portions **50a**, **50b** are selectively coupled during use of the loader, and selectively separable after the use of the loader so that the work vehicle may be driven away from the loader such as for storage of the loader or for performing other functions with the work vehicle or the like. The first and second portions **50a**, **50b** are also selectively coupleable prior to use of the loader so that the work vehicle may be driven toward the loader and couple with the loader such as for use of the loader for performing tasks with the work vehicle or the like. For this a dashed line is illustrated for designating that the first and second portions **50a**, **50b** may be selectively separated. In the example, one or the other of the first and second portions **50a**, **50b** may carry electronics and valves responsive to signals for selective connections to the hydraulic source **48** in response to the control signal **150**. It is to be appreciated that the portion of the multi-coupler device **50** carrying the electronics and valves may be in the first portion **50a** or the second portion **50b** or in both the first and second portions **50a**, **50b** and, further, the portion of the multi-coupler device **50** carrying the electronics and valves may be disposed on the loader **13** or on the tractor **10** as may be necessary or desired. It is further to be appreciated that the first portion **50a** of the multi-coupler device **50** remains with the associated work vehicle **1** and the second portion **50b** remains with the loader **13** when the associated work vehicle **1** is driven away from the loader for use elsewhere leaving the second portion **50b** of the multi-coupler device **50** with the loader **13** supported by the parking stand unit **40** such as shown in FIG. 2.

(26) The motion of the hydraulic lifting cylinders **24** and/or hydraulic pivoting cylinders **28** may be controlled in accordance with an aspect of the example embodiment by the attach assist apparatus **100** automatically and/or semi-automatically operating the multi-coupler device **50** in accordance with an embodiment while the front loader **13** is detached from the tractor **10** and while the first and second portions **50a**, **50b** may be selectively coupled. For example the attach assist apparatus **100** may function in an RTP mode to automatically return the mounting mast **18** of the front loader **13** to a position favorable to facilitate 1-step connection of the mounting mast **18** with the mounting frames **16** on the tractor **10**, thereby attaching the loader **13** with the tractor **10**. In addition, the mounting mast **18** of the front loader **13** may be returned to one or more pre-stored position(s) in response to operator input with the assistance of the attach assist apparatus **100** providing suitable instructions for helping to assist the operator in facilitating 1-step connection of the mounting mast **18** with the mounting frames **16** on the tractor **10**, thereby attaching the loader **13** with the tractor **10**.

(27) Actuation of the hydraulic pivoting cylinders **28** causing them to extend urges the bucket implement **23** to rotate relative to the carrier arms **21** in the direction A, thereby causing the mounting mast **18** to lower in the general direction A'. In this instance, the front loader essentially pivots counterclockwise about the foot **43**. Similarly, actuation of the hydraulic pivoting cylinders **28** causing them to retract urges the bucket implement **23** to rotate relative to the carrier arms **21** in the direction B, thereby causing the mounting mast **18** to raise in the general direction B'. In this instance, the front loader essentially pivots clockwise about the parking stand unit **40**.

(28) Actuation of the hydraulic lifting cylinders **24** causing them to extend urges the mast arms **19** to pivot clockwise relative to the carrier arms **21** as viewed in the drawing Figure, thereby causing the mounting mast **18** to raise in the general direction B'. Similarly, actuation of the hydraulic lifting cylinders **24** causing them to retract urges the mast arms **19** to pivot counterclockwise relative to the carrier arms **21** as viewed in the drawing Figure, thereby causing the mounting mast **18** to lower in the general direction A'.

(29) In addition, a set of one or more sensors **60** are disposed at one or more locations on the front loader **13**, wherein the one or more sensors **60** generate a corresponding set of one or more signals

70 that are representative of the orientation and/or configuration of the various components of the front loader **13**. In accordance with the example embodiments herein, the set of one or more signals **70** that are representative of the orientation and/or configuration of the various components of the front loader **13** and generated by the one or more sensors **60** may be received directly by the attach assist control unit **102** (FIG. 7) of the loader attach assist apparatus **100**, and/or they may be received indirectly by the attach assist control unit **102** of the loader attach assist apparatus **100** via the ECM **135** operatively coupled with the attach assist control unit **102** via a network link **130** (FIG. 7) or the like.

(30) For sensing the pivot angle or tilt angle of the front loader tool **22**, a tool inclination sensor **61** is provided which is operatively connected directly or indirectly with the pivoting linkage **27**. The tool inclination sensor **61** senses the pivot angle between the front loader tool **22** and the carrier arms **21**. Since the parking stand unit **40** is rigid and in an opened or deployed position as shown in FIG. 2, the signal **71** generated by the tool inclination sensor **61** is essentially representative of the inclination of the carrier arms **21** relative to the ground supporting the front loader **13**.

(31) A link inclination sensor **62** is disposed on the front loader in a position substantially as shown and is operatively coupled directly or indirectly with the link member **25** and/or the mast arms **19**. The link inclination sensor **62** senses the pivot angle between the link member **25** and the mast arms **19**, and generates a link inclination signal **72** that is representative of the sensed pivot angle between the link member **25** and the mast arms **19**.

(32) A mounting mast inclination sensor **63** is disposed on the front loader in a position substantially as shown and is operatively coupled directly or indirectly with the mounting mast members **18** and/or the mast arms **19**. The mounting mast inclination sensor **63** senses the pivot angle between the mounting mast members **18** and the mast arms **19**, and generates a mounting mast inclination signal **73** that is representative of the sensed pivot angle between the mounting mast members **18** and the mast arms **19**.

(33) A mast arm inclination sensor **64** is disposed on the front loader in a position substantially as shown and is operatively coupled directly or indirectly with one or more of the mast arms **19**. The mast arm inclination sensor **64** senses the inclination of the mast arms **19**, and generates a mast arm inclination signal **74** that is representative of the sensed inclination of the mast arms **19** relative to the ground supporting the front loader **13**.

(34) In the example embodiments as described herein, the sensors **61-64** may, for example, be magnetic sensors, potentiometric or optical rotational angle sensors, rotary sensors, rotation sensors, inclination sensors, or any other type of sensors for determining the relative orientations and positions of the various components of the front loader and for generating electrical signals representative of the determined relative positions. In that way, the attach assist apparatus **100** of the example embodiment may determine the position of the mounting mast **18** for helping to facilitate attachment of the loader **13** onto the tractor in an efficient manner for ease of operation.

(35) According to an aspect of the example embodiment, a front loader arrangement of the type described above is provided with an automated adjusting system and method, by means of which the hydraulic lifting cylinders **24** can be brought into a floating condition when the hydraulic fluid supply is decoupled. Owing to the fact that the hydraulic lifting cylinders **24** are brought into a floating position, the mounting masts **18** can move relatively freely and are not blocked by the hydraulic cylinders, as is otherwise customary during the coupling, since a hydraulic connection between the vehicle and front loader arrangement has been disconnected during the decoupling and, during the coupling, are selectively actuated automatically as may be necessary or desired by the attach assist apparatus **100** in one or more further steps if the mounting masts **18** fail to become coupled onto the mounting frames **16**. Each of the mounting frames **16** can, for example, comprise a ramp along which each of the mounting masts **18** may be guided and displaced for the coupling and decoupling. By means of the floating position, the mounting mast can be pivoted about the coupling point thereof with respect to the loader arm, i.e., relative to the loader arm, and, for

example during the coupling, can be adapted to a positioning angle with respect to the mounting frame or aligned with respect to the latter. An alignment takes place here by the bearing bolts of the mounting mast sliding over the ramp during the coupling and being guided into the catch hooks, wherein, by means of this interaction, the bearing bolts are brought into the position thereof provided for the coupling. Without the above-mentioned automated control over the floating position of the hydraulic lifting cylinders **24**, the mounting mast **18** would be in a rigid connection with respect to the loader arm such that the mounting mast **18** would not be able to carry out any pivoting movements relative to the loader mast arms **19**.

(36) Therefore, an initial step in accordance with an example embodiment is to pre-position the level and position or orientation of the mounting mast **18** relative to the ground supporting the front loader **13** prior to moving the tractor **10** towards the loader **13** for connection therewith should the geometrical dimensions between vehicle and front loader arrangement have changed during a time in which the front loader arrangement was decoupled from the vehicle, which is entirely possible, for example due to location instabilities (ground unevenness, subsidence of the ground, etc.), change in the filling of the tires of the vehicle, changes to the tire size, leakages in the hydraulics of the front loader device or because of other circumstances. Then, as a second step in accordance with an example embodiment and in the event that slight misalignment persists after the initial step of pre-positioning the level and position or orientation of the mounting mast **18** relative to the ground making coupling of the front loader arrangement considerably more difficult because of the rigid connection between mounting mast and loader arm, the attach assist apparatus **100** performs an automated control over the floating position of the loader by controlling directly and/or indirectly via the ECM of the tractor hydraulic control components **69** of the loader arrangement **11** including for example control over the hydraulic lifting cylinders **24**.

(37) FIG. **3** is a schematic side view of a mounting mast in cross section and of a mounting frame of the front loader arrangement of FIGS. **1** and **2** in a decoupled state, and FIG. **4** is a further schematic side view of the mounting mast of the mounting frame of FIGS. **1** to **3** in a coupled state. FIGS. **3** and **4** illustrate a respective combination of the mounting mast **18** and mounting frame **16** at the right side of the front loader arrangement **12** in enlarged form in a plurality of side views and in a plurality of cross-sectional views, in which further individual details of the front loader arrangement **12** are illustrated in detail.

(38) Each of the mountings frame **16** have a ramp **80** or ramp-shaped configuration that extends between the front and rear catch hooks **36** and **38**, and, in the lower region of the ramp, a depression or notch **81** is defined directly to the rear of the lower or front catch hook **36**.

(39) The mounting mast **18** comprises an upper bearing point and a lower bearing point respectively in the form of an upper bearing bolt **82** and a lower bearing bolt **83**. It is conceivable here for the combination of catch hooks **36**, **38** and bearing bolts **82**, **83** also to be able to be realized in a manner the other way around, such that the bearing bolts **82**, **83** are formed on the mounting frame **16** and the catch hooks **36**, **38** are formed on the mounting mast **18**. Furthermore, an upper pivoting bolt **84**, a central pivoting bolt **85** and a lower pivoting bolt **86** are arranged on the mounting mast **18**, with the pivoting bolts respectively pivotally connecting the tilting linkage **26**, the front loader arm **20** and the hydraulic lifting cylinder **24** to the mounting mast **18**.

(40) The front loader **13** or the mounting mast **18** is coupled to the mounting frame **16** as described below with reference to FIGS. **4** and **5**. The mounting frame **16** is moved from a decoupled state, in which the front loader **13** is set down and is separated hydraulically and mechanically from the tractor **10**, in the direction of the mounting mast **18** corresponding to the movement indicated by the arrow V (FIG. **3**). In an example embodiment the movement of the mounting frame **16** in the direction of the mounting mast **18** corresponding to the movement indicated by the arrow V is in response to the attach assist apparatus **100** generating a signal for developing visual command that instruct an operator of the tractor **10** to drive the tractor forward. In another example embodiment the movement of the mounting frame **16** in the direction of the mounting mast **18** corresponding to

the movement indicated by the arrow **V** is in response to the attach assist apparatus **100** generating a signal for delivery to one or more system(s) of the tractor **10** to automatically drive the tractor forward without the need for operator intervention. The upper bearing bolt **82** begins to enter into engagement with the ramp **80** and slides up along the inclined plane of the ramp **80** and along the ramp surface until the bearing bolt enters into engagement with the upper catch hook **38**. At this same time, the lower bearing bolt **83** enters into engagement with the lower catch hook **36**. Approximately at the time at which the bearing bolts **82**, **83** enter into engagement with the catch hooks **38**, **36**, the lower catch hook **36**, or a lower end of the ramp **80**, abuts against a supporting means **90** and begins to pivot the latter counter to the pre-tensioning thereof from the position thereof, which is oriented substantially perpendicularly to the ramp surface and in which the supporting means takes up the supporting position, into an increasingly deflected position. When the supporting position of the supporting means **90** is cancelled, a spring **91** on a pivoting locking bar **92** causes the latter to move with a lower pivoting arm **93** thereof in the direction of the ramp surface such that, finally, a step **94** of the lower pivoting arm **93** comes to bear against the ramp **80** and slides up along the ramp. As can best be seen in FIG. 4, the mounting mast **18** may reach an end position relative to the mounting frame **16**, at which end position the bearing bolts **82**, **83** come to bear completely in the catch hooks **38**, **36**. At the same time, the spring pre-tensioned pivoting locking bar **92** takes up the locking position thereof and latches with the step **94** formed on the first pivoting arm **93** in the depression **81** such that the mounting mast **18** is locked to the mounting frame **16**, since relative movement between mounting mast **18** and mounting frame **16** is now blocked firstly by the catch hooks and secondly by the step **94** which has come to bear in the depression **81**. The front loader is now in the operating position and can be put into operation.

(41) A loader locked sensor **65** is disposed on or near the interface between the mounting mast **18** and the mounting frame **16** in a position substantially as shown and is operatively coupled directly or indirectly with the attach assist apparatus **100** of the example embodiment. The loader locked sensor **65** senses that the mounting mast **18** is fully and completely locked to or with the mounting frame **16**, and generates a loader locked signal **75** that is representative of the sensed mutually locked condition of the mounting mast **18** and the mounting frame **16**.

(42) In the example embodiments as described herein, with the loader and the associated work vehicle being in mutual contact, a coupled confirmation signal is received such as for example from the loader locked sensor **65**, wherein the coupled confirmation signal is representative of a confirmation of the loader being coupled with the associated work vehicle. The loader and the associated work vehicle being in mutual contact may occur, for example, following or subsequent to the movement of the associated work vehicle towards the loader. In the method and also with the loader and the associated work vehicle being in mutual contact, a forward movement pause instruction signal is generated responsive to receiving the coupled confirmation signal such as for example from the loader locked sensor **65**, wherein the forward movement pause instruction signal is generated for pausing the movement of the associated work vehicle towards the loader.

(43) In the example embodiments as described herein, the loader locked sensor **65** may, for example, comprise magnetic sensors, potentiometric or optical rotational angle sensors, rotary sensors, rotation sensors, inclination sensors, or any other type of sensors for determining the that the mounting mast **18** is fully and completely locked to or with the mounting frame **16** and for generating electrical signals representative of the determined locked condition. In that way, the attach assist apparatus **100** of the example embodiment may determine that the mounting mast **18** is fully and completely locked to or with the mounting frame **16** for helping to facilitate attachment of the loader **13** onto the tractor in an efficient manner for ease of operation.

(44) An automated hydraulic adjusting system can be provided in accordance with an example embodiment and as shown for example in FIG. 5 which comprises a hydraulic accumulator and an automatically actuated shut-off valve via which the hydraulic cylinder can be connected fluidly to the hydraulic accumulator. By selectively opening of the shut-off valve under the control of the

attach assist apparatus **100** of the example embodiments, the hydraulic accumulator can be fluidly connected to a head end chamber of a hydraulic cylinder acting on one side (the head end chamber constituting the lifting chamber), and therefore hydraulic oil can flow out of the hydraulic cylinder into the hydraulic accumulator and vice versa and the piston in the hydraulic cylinder and the piston rod connected to the piston can move freely. As a result, the mounting mast, as already described above, can also move relatively freely or can be freely pivoted or aligned. The shut-off valve is preferably opened directly after the front loader is decoupled from the vehicle, with a hydraulic oil supply from the vehicle for the hydraulic cylinder of the front loader arrangement (operating hydraulics) customarily also being interrupted or decoupled, as a result of which the hydraulic cylinder is already held in a floating position and is prepared for the next coupling. Of course, the shut-off valve can also be opened just immediately before the next coupling. An automated opening under the control of the attach assist apparatus **100** can be provided here, which may ensure that the floating position for the hydraulic cylinder can be set independently of the vehicle hydraulics or of operating hydraulics for the front loader arrangement. It is entirely also conceivable here to provide an automated mechanical and/or an automated electric device for opening the shut-off valve if the latter can be operated independently of the operating hydraulics of the front loader or ensures the opening of the shut-off valve independently of operating hydraulics, i.e. enables the shut-off valve to open if a connection to the operating hydraulics is interrupted.

(45) In addition to the above-mentioned design of the hydraulic cylinder, the automated hydraulic adjusting system can also be formed as shown for example in FIG. **6** in a double-acting manner with rod end and head end chambers, wherein the shut-off valve is arranged between the two chambers and between the hydraulic accumulator and the head end chamber. By opening of the shut-off valve, the two chambers are interconnected, with a floating position for the hydraulic cylinder being achieved, and with the volumetric differences or the cross-sectional differences between the two chambers being compensated for by the hydraulic accumulator. As a result, the mounting mast can here also, as described above, move relatively freely or can be freely pivoted or aligned. The shut-off valve is here too preferably opened directly after the decoupling of the front loader from the vehicle, as a result of which the hydraulic cylinder is already held in the floating position and is ready for the next coupling. Of course, the shut-off valve can also be opened just immediately before the next coupling. An automated opening under the control of the attach assist apparatus **100** can be provided here, which may ensure that the floating position for the hydraulic cylinder can be set independently of the vehicle hydraulics or of operating hydraulics for the front loader arrangement. It is entirely also conceivable here to provide an automated mechanical and/or an automated electric device for opening the shut-off valve if the latter can be operated independently of the operating hydraulics of the front loader or ensures the opening of the shut-off valve independently of operating hydraulics, i.e. enables the shut-off valve to open if a connection to the operating hydraulics is interrupted.

(46) In particular and in order to make coupling easier, it is expedient, as has already been described, that, during the coupling, the mounting mast **18** can be adapted as freely as possible to the position of the mounting frame **16** such that the bearing bolts **82**, **83** are guided as exactly and precisely as possible into the catch hooks. This is achieved in that the rigid connection between the mounting mast **18** and front loader arms **20**, in which, when the hydraulic supply is disconnected, the lifting cylinders **24** remain in the position thereof which they have thus taken up and hold the mounting mast **18** rigidly with respect to the loader arm **20**, is cancelled. The mounting mast **18** is thereby brought into a movable state in which the mounting mast can be pivoted relative to the loader arm **20**. For this purpose, as FIGS. **5** and **6** schematically show, a shut-off valve **52** and a hydraulic accumulator **53** are provided, wherein FIG. **5** illustrates a hydraulic circuit for a single-acting lifting cylinder and FIG. **6** illustrates a double-acting lifting cylinder. In both cases, the shut-off valve **52** is designed to be automatically selectively actuated by the attach assist apparatus **100** of the example embodiments during the coupling and/or during the decoupling of the front loader

13, when the hydraulic supply is connected or disconnected. In the example embodiments and in both cases of the single-acting and double-acting hydraulic adjusting systems, the shut-off valve **52** is responsive to a control signal **52'** generated by the attach assist apparatus **100** to automatically selectively open in response to receiving the control signal **52'**. In accordance with the example embodiments herein, the control signal **52'** may be generated directly by the attach assist control unit **102** (FIG. 7) of the loader attach assist apparatus **100**, and/or the control signal **52'** may be generated indirectly by the attach assist control unit **102** via the ECM **135** operatively coupled with the attach assist control unit **102** via a network link **130** or the like.

(47) With regard to FIG. 5, part of a hydraulic system **30** for the supply of two lifting cylinders **24** each acting on one side of the tractor is depicted, the lifting cylinders having a piston **31** and being hydraulically supplied via a hydraulic supply line **32**, wherein the supply line **32** extends from a coupling part **33** of a hydraulic quick coupler (not shown) to head end chambers **34** of the lifting cylinders **24**. As can be seen in FIG. 1, the head end of the lifting cylinder **24** on the right side of the tractor **10** is connected to the loader arm **20** and on the rod end is connected to the mounting mast **18**, with it being understood that the lifting cylinder **24** on the left side of the tractor is similarly connected, and therefore the loader arm **20** can be raised by hydraulic filling of the head end chambers **34**. The loader arm **20** is lowered here because of gravitational force, wherein a control valve (not shown) of the hydraulic system **30** is correspondingly actuated in order to raise and lower the front loader, and the hydraulic oil can be pressed out of the head end chambers **34** by the dead weight of the front loader. The two lifting cylinders **24** are furthermore connected to each other by a connecting line **54** and are connected to the supply line **32**, and therefore the two lifting cylinders are pressurized uniformly. The shut-off valve **52** is arranged in a further connecting line **55** which connects the supply line **32** to the hydraulic accumulator **53**. By opening of the shut-off valve **52**, it is ensured that, when the hydraulic supply is disconnected, the hydraulic fluid is not trapped in the head end chambers **34**, but rather can escape into the hydraulic accumulator **53**, and vice versa. As a result, the lifting cylinders **24** can take up a floating movement of the mounting mast **18** in relation to the loader arm **20**, and therefore, when the shut-off valve **52** is open, the mounting mast **18** can be aligned in a freely pivotable manner during the coupling. The opening of the shut-off valve **52** already ideally takes place during the decoupling of the front loader **13** and can thus be carried out by an operator in one working sequence together with an unlocking of the pivoting locking bar **92** described above and the separating of the hydraulic supply (separating of the quick coupler **50**). Owing to the fact that the opening of the shut-off valve **52** already takes place during decoupling of the front loader **13**, the latter is already prepared for the next coupling operation, and therefore an operator can couple the front loader **13** to the freely movable mounting mast **18** without leaving the tractor **10**. After the coupling of the front loader **13**, the shut-off valve **52** can be closed for the starting up of the hydraulic function (connecting of the hydraulic supply) in order to ensure a hydraulic operation of the front loader **13**. This can likewise take place in one working sequence together with the connection of the quick coupler **50**.

(48) By operating the shut-off valve **52** to selectively open and close in response to the control signal **52'** generated by the attach assist apparatus of the example embodiments while the tractor is being moved forward, it is ensured that, when the hydraulic supply is alternately disconnected and connected, the hydraulic fluid is not constantly trapped in the head end chambers **34**, but rather can alternately escape into the hydraulic accumulator **53**, and vice versa. As a result, the lifting cylinders **24** can take up alternating floating and locked movements of the mounting mast **18** in relation to the loader arm **20** and, therefore, when the shut-off valve **52** is open the mounting mast **18** can be aligned in a freely pivotable manner during the coupling, and when the shut-off valve **52** is closed the mounting mast **18** can become locked in place relative to the other components of the front loader and also relative to components of the mounting frame including the catch hooks **36**, **38** to establish a resetting or re-initialization of the alignment procedure. The alternate opening and closing of the shut-off valve **52** as controlled by the attach assist apparatus **100** of the example

embodiments **100** ideally takes place during the coupling of the front loader **13** when the loader locked sensor **65** does not sense that the mounting mast **18** is fully and completely locked to or with the mounting frame **16**, and when the loader locked signal **75** representative of the sensed mutually locked condition of the mounting mast **18** and the mounting frame **16** is not received by the attach assist apparatus **100** within a predetermined time during the coupling of the front loader **13**. In this way, the coupling of the front loader **13** can be carried out in one working sequence (“1-trip”) by the automated assistance of the attach assist apparatus **100**. In the above it is to be appreciated that in accordance with the example embodiments herein, the control signal **52'** may be generated directly by the attach assist control unit **102** (FIG. 7) of the loader attach assist apparatus **100** to perform the alternate opening and closing of the shut-off valve **52**, and/or the control signal **52'** may be generated indirectly by the attach assist control unit **102** via the ECM **135** operatively coupled with the attach assist control unit **102** via a network link **130** or the like to perform the alternate opening and closing of the shut-off valve **52**.

(49) With regard to FIG. 6, a part of a hydraulic system **30** for the hydraulic supply of two double-acting lifting cylinders **24** is depicted, the lifting cylinders having a piston **31** and by hydraulically supplied via a hydraulic supply line **32** and a hydraulic supply line **87**. The supply line **32** extends from a coupling part **33** of a hydraulic quick coupler (not shown) to head end chambers **34** of the lifting cylinders **24**, wherein a connecting line **89** extends from the coupling part **33** to rod end chambers **88**. As with the single-acting lifting cylinders **24** described above, the double-acting lifting cylinders **24** at the opposite sides of the tractor **10** have their head ends connected to the loader arm **20** and their rod ends respectively coupled to the masts **18** at the opposite sides of the tractor such that the loader arm **20** is raised and lowered by hydraulic filling of the head end chambers **34**, wherein a corresponding activation of a control valve (not shown) of the hydraulic system **30** takes place in a known manner for lifting and lowering the front loader. The head end chambers **34** of the two lifting cylinders **24** are connected to each other by a connecting line **54** that is connected to the supply line **32**. Furthermore, the rod end chambers **88** of the lifting cylinders **24** are connected to each other by a connecting line **89** that is connected to the supply line **87**. As a result, the two hydraulic cylinders **24** are pressurized uniformly. The hydraulic accumulator **53** is connected to the supply line **87** via a connecting line **59** and therefore also to the connecting line **89** and to the rod end chambers **88**. The shut-off valve **52** is arranged in a line **58** coupled between the supply line **32** and the connecting line **59**.

(50) By automatically selectively opening the shut-off valve **52** by the attach assist apparatus **100** of the example embodiments, it is ensured that, when the hydraulic supply is disconnected, the hydraulic fluid is not trapped in the chambers **34**, **88**, but rather can escape into the hydraulic accumulator **53** and into the respective other chamber **34**, **88**, and vice versa. As a result, the lifting cylinders **24** can take up a floating position in which the pistons **31** are freely movable, wherein volume differences between the head end chambers **34** and the rod end chambers **88** are compensated for by the hydraulic accumulator **53**. This state, in turn, permits a relatively free movement of the mounting masts **18** in relation to the loader arm **20**, and therefore, when the shut-off valve **52** is open, the mounting masts **18** can be aligned in a freely pivotable manner during the coupling. The opening of the shut-off valve **52** during decoupling of the front loader **13** can take place here in one working sequence together with unlocking of the pivoting locking bar **92** and the separating of the hydraulic supply (separating the quick coupler) by an operator. Owing to the fact that the shut-off valve **52** is already opened during the decoupling of the front loader **13**, the latter is also already prepared for the next coupling operation, and therefore an operator can couple the front loader **13** to the freely movable mounting masts **18** without leaving the tractor **10**. The shut-off valve **52** also has to be closed here, after the coupling of the front loader **13**, for starting up of the hydraulic function (connecting of the hydraulic supply), in order to ensure a hydraulic operation of the front loader **13**. However, this can likewise take place in one working sequence together with the connection of the quick coupler.

(51) By operating the shut-off valve **52** to selectively open and close in response to the control signal **52'** generated by the attach assist apparatus of the example embodiments while the tractor is being moved forward, it is ensured that, when the hydraulic supply is alternately disconnected and connected, the hydraulic fluid is not constantly trapped in the head end chambers **34**, but rather can alternately escape into the hydraulic accumulator **53**, and vice versa. As a result, the lifting cylinders **24** can take up alternating floating and locked movements of the mounting mast **18** in relation to the loader arm **20** and, therefore, when the shut-off valve **52** is open the mounting mast **18** can be aligned in a freely pivotable manner during the coupling, and when the shut-off valve **52** is closed the mounting mast **18** can become locked in place relative to the other components of the front loader and also relative to components of the mounting frame including the catch hooks **36**, **38** to establish a resetting or re-initialization of the alignment procedure. The alternate opening and closing of the shut-off valve **52** as controlled by the attach assist apparatus **100** of the example embodiments **100** ideally takes place during the coupling of the front loader **13** when the loader locked sensor **65** does not sense that the mounting mast **18** is fully and completely locked to or with the mounting frame **16**, and when the loader locked signal **75** representative of the sensed mutually locked condition of the mounting mast **18** and the mounting frame **16** is not received by the attach assist apparatus **100** within a predetermined time during the coupling of the front loader **13**. In this way, the coupling of the front loader **13** can be carried out in one working sequence ("1-trip") by the automated assistance of the attach assist apparatus **100**.

(52) FIG. 7 is a block diagram that illustrates a representative attach assist apparatus **100** according to the example embodiments. The attach assist apparatus **100** is suitable for executing embodiments of one or more software systems or logic modules that perform methods for assisting an operator of an associated work vehicle to couple a loader onto the work vehicle.

(53) The representative example attach assist apparatus **100** includes an attach assist control unit **102**, a set of operator interface devices **104**, a set of loader interface devices **106**, and a set of tractor interface devices **108**. The attach assist control unit **102** of the attach assist apparatus **100** according to the illustrated embodiment includes a bus **110** or other communication mechanism for communicating information, a processor device **111** coupled with the bus **110** for processing information, and a non-transient memory device **112** configured to store data, information and/or instructions that are executable by the processor device **111**.

(54) The attach assist control unit **102** further includes a non-transient memory device **112** that may be of any memory type and also may include different memory types or portions including for example a random access memory (RAM) **113** or other dynamic storage device for storing information and/or instructions to be executed by the processor device **111**, and read only memory (ROM) **114** or other static storage device for storing static information and/or instructions for the processor device **111**.

(55) In accordance with the descriptions herein, the terms "computer-readable medium" and "memory device" as used herein refer to any non-transitory media that participates in providing logic instructions and/or data to the processor device **111** for execution and/or processing. Such a non-transitory medium may take many forms, including but not limited to volatile and non-volatile media. Non-volatile media includes, for example, optical or magnetic disks. Volatile media includes dynamic memory for example and does not include transitory signals, carrier waves, or the like. Common forms of computer-readable media include, for example, a floppy disk, a flexible disk, hard disk, magnetic tape, or any other magnetic medium, a CD-ROM, any other optical medium, punch cards, papertape, any other physical medium with patterns of holes, a RAM, PROM, and EPROM, a FLASH-EPROM, any other memory chip or cartridge, or any other tangible non-transitory medium from which a computer can read.

(56) In addition and further in accordance with the descriptions herein, the term "logic", as used herein with respect to the Figures, includes hardware, firmware, software in execution on a machine, and/or combinations of each to perform a function(s) or an action(s), and/or to cause a

function or action from another logic, method, and/or system. Logic may include a software controlled microprocessor, a discrete logic (e.g., ASIC), an analog circuit, a digital circuit, a programmed logic device, a memory device containing instructions, and so on. Logic may include one or more gates, combinations of gates, or other circuit components.

(57) With continued reference to FIG. 7, the non-transient memory device **112** may suitably store attach assist control logic **116** comprising instructions for execution by the processor device **111** for controlling one or more portions of the loader **13** and/or one or more portions of the tractor **13** for assisting an operator latch a loader onto the tractor. The non-transient memory device **112** may further suitably store attach assist data **118** comprising information such stored sensor level data relating to desired loader positions for RTP operation by way of example. In this regard, the attach assist apparatus **100** may function in the RTP mode to automatically return the mounting mast **18** of the front loader **13** to a position favorable to facilitate 1-step connection of the mounting mast **18** with the mounting frames **16** on the tractor **10**, thereby attaching the loader **13** with the tractor **10**. The mounting mast **18** of the front loader **13** may be returned to one or more positions stored as attach assist data **118** in the non-transient memory device **112** for helping to assist the operator in facilitating 1-step connection of the mounting mast **18** with the mounting frames **16** on the tractor **10**, thereby attaching the loader **13** with the tractor **10**. The attach assist data **118** may further comprise data relating to images or the like for display to a human operator providing instructions or the like for assisting in efficiently attaching the loader onto the tractor.

(58) In addition to the above, the attach assist control unit **102** of the example embodiment further includes a communication interface **119** coupled with the bus **110**. The communication interface **119** provides in the example embodiment a two-way data communication coupling between the attach assist control unit **102** of the attach assist apparatus **100** and a network link **130** to be described below in connection with the set of tractor interface devices **108**. The communication interface **119** provides a connection between the attach assist control unit **102** and a local network **132** such as for example a local network of the work vehicle **10** such as a Controller Area Network (CAN) bus or the like. The attach assist apparatus **100** may use the communication interface **119** and the network link **130** to indirectly control the various hydraulic control components **69** of the loader arrangement **11** by sending and/or receiving signals and/or commands between the attach assist control unit **102** of the loader attach assist apparatus **100** and the ECM **135** native to the associated work vehicle **1** via a communication link or the like for moving the cylinders **24**, **28** by the ECM **135** executing the signals and/or commands and thereby effecting the desired movement of the loader arrangement **11**. The communication interface **119** may be a CAN card to provide a data communication connection to a compatible CAN bus. As another example, communication interface **119** may be a local area network (LAN) card to provide a data communication connection to a compatible LAN. For example, communication interface **119** may be an integrated services digital network (ISDN) card or a modem to provide a data communication connection to a corresponding type of telephone line. Wireless links may also be implemented. In any such implementation, communication interface **119** may be a wireless receiver/transmitter, i.e. a transceiver operable to send and receive electrical, electromagnetic, radio frequency (RF), and/or optical signals that carry data streams such as digital data streams representing various types of information.

(59) The set of operator interface devices **104** of the attach assist control unit **102** of the example attach assist apparatus **100** may include several subsystems or modules to perform various interactions with the human operator as set forth herein. A benefit of the subject application is to provide improved guidance by displaying guidance and other instructions on a screen **121** of a display unit **120** that is viewable from an operator's seat of the associated work vehicle for helping the operator to view current positions of the loader and tool mechanisms as well as to visualize preferred or recommended instructions from the attach assist control unit **102** for movement of the loader and tool mechanisms and optionally of the tractor to safely and efficiently couple the loader

with the tractor. An output device **124** may also be provided such as in the form of a sound generating device such as a speaker to help improve guidance assistance by generating audible guidance instructions in the form of audible instructions and/or suitable instructional noises such as beeps, voice messages or the like that can be heard from an operator's seat of the associated work vehicle for helping the operator to be alerted to current positions of the mechanisms as well as to be audibly instructed of preferred or recommended paths for movement of the loader and tool mechanisms or of the tractor to safely and efficiently attach the loader **13** onto the tractor **10**.

(60) The example embodiment of the attach assist apparatus **100** further includes an input device **125** operatively coupled with the attach assist control unit **102**. The input device **125** may be used during a use of the attach assist apparatus **100** for providing assistance in coupling a loader with an associated work vehicle. The input device **125** may also be used during a training of the attach assist apparatus **100** for selection of received one or more loader position training signals such as for example signals **71-74** received from the set **60** of sensors **61-64** representative of desired positions of the mast and carrier arms **19, 21** for alignment and orientation and positioning of the loader **13** in a pre-attachment position. In an example embodiment the input device **125** may include a touchscreen portion **122** of the display unit **120**, a pointer device operatively coupled with the attach assist control unit **102**, or any other device or means of communicating training and other information to the attach assist control unit **102** of the attach assist apparatus **100** such as for example for receiving training signals and/or other information signals by the attach assist control unit **102** of the attach assist apparatus **100**.

(61) The set of loader interface devices **106** of the attach assist control unit **102** of the example attach assist apparatus **100** may include several subsystems or modules to perform various interactions with the loader **13** as set forth herein. By way of example, a sensor input unit **126** of the set of loader interface devices **106** is configured to receive into the attach assist control unit **102** of the example attach assist apparatus **100** signals **70** generated by the set of one or more sensors **60** are disposed at one or more locations on the front loader **13**. In addition, a control signal output unit **127** of the set of loader interface devices **106** is configured to deliver from the attach assist control unit **102** of the example attach assist apparatus **100** loader control signals **68** generated by the attach assist apparatus **100**, wherein the loader control signals **68** are configured to operate the various hydraulic control components **69** of the loader arrangement **11** for moving the various hydraulic cylinders and the like thereby effecting the desired movement of the loader arrangement **11**.

(62) By way of example, a sensor input unit **126** of the set of loader interface devices **106** is configured to receive into the attach assist control unit **102** of the example attach assist apparatus **100** the signal **71** generated by the tool inclination sensor **61** that is representative of the inclination of the carrier arms **21** relative to the ground supporting the front loader **13**. The sensor input unit **126** is also configured to receive the link inclination signal **72** into the attach assist control unit **102** of the example attach assist apparatus **100** from the link inclination sensor **62**, wherein the link inclination signal **72** is representative of the sensed pivot angle between the link member **25** and the mast arms **19**. The sensor input unit **126** is further configured to receive the mounting mast inclination signal **73** into the attach assist control unit **102** of the example attach assist apparatus **100** from the mounting mast inclination sensor **63**, wherein the mounting mast inclination signal **73** is representative of the sensed pivot angle between the mounting mast members **18** and the mast arms **19**. The sensor input unit **126** is still further configured to receive the mast arm inclination signal **74** into the attach assist control unit **102** of the example attach assist apparatus **100** from the mast arm inclination sensor **64**, wherein the mast arm inclination signal **74** is representative of the sensed inclination of the mast arms **19** relative to the ground supporting the front loader **13**. The sensor input unit **126** is yet still further configured to receive the loader locked signal **75** into the attach assist control unit **102** of the example attach assist apparatus **100** from the loader locked sensor **65**, wherein the loader locked signal **75** is representative of the sensed mutually locked

condition of the mounting mast **18** and the mounting frame **16**.

(63) The set of tractor interface devices **108** of the attach assist control unit **102** of the example attach assist apparatus **100** may include several subsystems or modules to perform various communication and/or command interactions between the tractor **10** and the attach assist apparatus **100** as set forth herein. By way of example, the tractor interface devices **108** includes a network link **130** that typically provides data communication through one or more networks to other data devices. For example, network link **130** may provide a connection through a local network **132** to a diagnostic host computer (not shown) of the like for supporting configuration of the system as desired or necessary. An Internet Service Provider (ISP) **133** may provide data communication services indirectly through the Internet via the network **132** or directly through the network link **130**.

(64) The example attach assist apparatus **100** can send messages and receive data, including program code, through the network(s), network link **130** and communication interface **119**. In the Internet-connected example embodiment, the attach assist apparatus **100** is operatively connected with a plurality of external public, private, governmental or commercial servers (not shown) configured to execute a web application in accordance with example embodiments.

(65) In an example embodiment during system training, the operator may first position the mast arms **19** and/or the carrier arms **21** of the front loader **13** to a desired position relative to the vehicle and/or relative to the ground, then click or drag and drop a virtual indicia on the touchscreen **122** using a pointer or finger onto a selected portion of the image of boom on the screen **121** while the boom is in the desired position, wherein the virtual indicia comprises the boom position training signal representative of a selectable target location for alignment of the mounting mast **18** relative to the mounting frame **16** to establish a desired relative physical position between the mounting mast **18** portion of the loader **13** and the mounting frame **16** attached with the tractor, or establish a desired relative physical position between the mounting mast **18** portion of the loader **13** relative to the ground supporting the loader.

(66) In an example embodiment, data may be saved into the non-transient memory device **112** during the training wherein the data is representative of the one or more signals **71-74** received from the one or more sensor devices **61-64** wherein the signals **71-74** are in turn representative of the desired position of the mast arms **19** and/or the carrier arms **21** of the front loader **13** relative to the vehicle and/or relative to the ground for pre-positioning the front loader **13** prior to moving the tractor **10** towards the loader **13** for connection therewith.

(67) In particular, data is stored into the non-transient memory device **112** during training that is representative of the signal **71** generated by the tool inclination sensor **61** that is in turn representative of the inclination of the carrier arms **21** relative to the ground supporting the front loader **13**.

(68) Further data may be stored into the non-transient memory device **112** during the training that is representative of the link inclination signal **72** from the link inclination sensor **62** wherein the link inclination signal **72** is in turn representative of the sensed pivot angle between the link member **25** and the mast arms **19**.

(69) Further data may be stored into the non-transient memory device **112** during the training that is representative of the mounting mast inclination signal **73** from the mounting mast inclination sensor **63** wherein the mounting mast inclination signal **73** is in turn representative of the sensed pivot angle between the mounting mast members **18** and the mast arms **19**.

(70) Further data may be stored into the non-transient memory device **112** during the training that is representative of the mast arm inclination signal **74** from the mast arm inclination sensor **64** wherein the mast arm inclination signal **74** is in turn representative of the sensed inclination of the mast arms **19** relative to the ground supporting the front loader **13**.

(71) In accordance with an example embodiment, the apparatus **100** includes an attach assist control unit **102** including a processor device **111**, a non-transient memory device **112** operatively

coupled with the processor device, and attach assist control logic **116** stored in the memory device. In the example embodiment, the processor device is operable to execute the attach assist control logic to generate a pre-position instruction signal for pre-positioning the loader in a predetermined orientation to facilitate the coupling. The processor device is further operable to execute the attach assist control logic to generate a forward movement instruction signal for initiating movement of the associated work vehicle towards the loader. The processor device is still further operable to execute the attach assist control logic to receive a coupled confirmation signal representative of a confirmation of the loader being coupled with the associated work vehicle. The processor device is yet still further operable to execute the attach assist control logic to generate based on receiving the coupled confirmation signal a forward movement pause instruction signal for pausing the movement of the associated work vehicle **1** towards the loader.

(72) The attach assist apparatus of the example embodiment further includes a human readable display unit operably coupled with the attach assist control unit, and an input device operably coupled with the attach assist control unit. The human readable display unit includes a screen operable to display images that are viewable by an operator of the associated work vehicle.

(73) The processor device of the example embodiment is operable to execute the attach assist control logic to generate the pre-position instruction signal by generating a pre-position instruction image and displaying the pre-position instruction image on the screen of the human readable display unit **120**, wherein the pre-position instruction image informs an operator of the associated vehicle how to pre-position the loader in the predetermined orientation to facilitate the coupling. The processor device of the example embodiment is operable to execute the attach assist control logic to generate the forward movement instruction signal by generating a forward movement instruction image, and displaying the forward movement instruction image on the screen of the human readable display unit, wherein the forward movement instruction image instructs the operator of the associated work vehicle to initiate the movement of the associated work vehicle towards the loader. The processor device of the example embodiment is operable to execute the attach assist control logic to receive the coupled confirmation signal by receiving a manual coupled confirmation signal by the input device from the operator of the associated vehicle, the coupled confirmation signal confirming by the operator that the loader is coupled with the associated work vehicle. The processor device of the example embodiment is operable to execute the attach assist control logic to generate the forward movement pause instruction signal by generating, responsive to receiving the coupled confirmation signal, a forward movement pause instruction image and displaying the forward movement pause instruction image on the screen of the human readable display unit, wherein the forward movement pause instruction image instructs the operator of the associated work vehicle to pause the movement of the associated work vehicle towards the loader.

(74) The processor device **111** of the attach assist apparatus **100** is operable to execute the attach assist control logic **116** to display an orientation image on the screen **121** of the human readable display unit **120**, wherein the orientation image is representative of desired orientations of one or more components of the loader informing the operator of the associated vehicle how to move the one or more components of the loader to predetermined positions for pre-positioning the loader in the predetermined orientation to facilitate the coupling.

(75) The processor device **111** of the attach assist apparatus **100** is operable to execute the attach assist control logic **116** to display the orientation image on the screen **121** of the human readable display unit **120** as one or more of an orientation of a carrier arm **21** of the loader **13**, an orientation of a mast arm **19** of the loader **13**, and/or an orientation of a mounting mast **18** of the loader **13** that is operably coupleable with a corresponding mounting frame **16** affixed to the associated work vehicle **1**.

(76) In the example embodiment herein, the attach assist apparatus **100** further includes a sensor input unit **126** operably coupled with the attach assist control unit **102**, and a loader locked sensor **65** operably coupled with the attach assist control unit **102** by the sensor input unit **126**, wherein

the loader locked sensor **65** is disposed at an interface between a mounting mast **18** of the loader **13** that is operably coupleable with a corresponding mounting frame **16** affixed to the associated work vehicle **1**. The processor device **111** of the attach assist apparatus **100** is operable to execute the attach assist control logic **116** to receive the coupled confirmation signal by receiving a loader locked signal **75** from the loader locked sensor **65** as the coupled confirmation signal representative of the confirmation of the loader **13** being coupled with the associated work vehicle **1**.

(77) In the example embodiment herein, the processor device **111** of the attach assist apparatus **100** is operable to execute the attach assist control logic **116** to generate a float condition instruction signal to establish a float condition in hydraulic lifting cylinders **24** of the loader **13** to relax a mounting mast **18** of the loader **13** that is operably coupleable with a corresponding mounting frame **16** affixed to the associated work vehicle **1** permitting the mounting mast **18** to move relative to the mounting frame **16** during the coupling.

(78) In the example embodiment herein, the processor device **111** of the attach assist apparatus **100** is operable to execute the attach assist control logic **116** to generate the float condition instruction signal by generating a float condition cycle instruction image on the screen **121** of the human readable display unit **120**, wherein the float condition cycle instruction image instructs the operator of the associated work vehicle to intermittently establish, with the loader **13** and the associated work vehicle **1** being in mutual contact and with the associated work vehicle **1** moving towards the loader **13**, the float condition between periods of a locked condition of the mounting mast **18** of the loader **13** wherein in the locked condition movement between the mounting mast **18** and the mounting frame **16** is prevented.

(79) In the example embodiment herein, the attach assist apparatus **100** further includes a control signal output unit **127** operably coupled with the attach assist control unit **102**, and a valve **52** operably coupled with the attach assist control unit **102** by the control signal output unit **127**, wherein the valve **52** is responsive to a control signal **52'** generated by the attach assist apparatus **100** to actuate to selectively relieve hydraulic fluid from a chamber of one or more hydraulic lifting cylinders **24** of the loader **13**. The processor device **111** is operable to execute the attach assist control logic **116** to generate the float condition instruction signal as the control signal **52'** to establish a float condition in hydraulic lifting cylinders **24** of the loader **13** to relax a mounting mast **18** of the loader **13** that is operably coupleable with a corresponding mounting frame **16** affixed to the associated work vehicle **1** permitting the mounting mast **18** to move relative to the mounting frame **16** during the coupling.

(80) In the example embodiment herein, the attach assist apparatus **100** further includes a communication interface **119** operably coupling the attach assist control unit **102** with a controller **135** of the associated work vehicle, and a loader locked sensor **65** operably coupled with the attach assist control unit **102** by the sensor input unit **126**, wherein the loader locked sensor **65** is disposed at an interface between a mounting mast **18** of the loader **13** that is operably coupleable with a corresponding mounting frame **16** affixed to the associated work vehicle **1**. The processor device **111** is operable to execute the attach assist control logic **116** to generate the pre-position instruction signal by delivering the pre-position instruction signal via the communication interface **119** to the controller **135** of the associated work vehicle to automatically pre-position the loader in the predetermined orientation to facilitate the coupling.

(81) In the example embodiment herein, the processor device **111** is operable to execute the attach assist control logic **116** to generate the forward movement instruction signal by delivering the pre-position instruction signal via the communication interface **119** to the controller **135** of the associated work vehicle to automatically initiate the movement of the associated work vehicle **1** towards the loader **13**.

(82) In the example embodiment herein, the processor device **111** is operable to execute the attach assist control logic **116** to receive the coupled confirmation signal by automatically receiving the coupled confirmation signal as a loader locked signal **75** from the loader locked sensor **65**, the

loader locked signal **75** from the loader locked sensor **65** being representative of the confirmation of the loader **13** being coupled with the associated work vehicle **1**.

(83) In the example embodiment herein, the processor device **111** is operable to execute the attach assist control logic **116** to generate the forward movement pause instruction signal by delivering, responsive to receiving the coupled confirmation signal, the forward movement pause instruction signal via the communication interface **119** to the controller the associated work vehicle to automatically pause the movement of the associated work vehicle **1** towards the loader **13**.

(84) FIG. **8** is a flow diagram illustrating a method **800** of assisting attaching a loader onto a work vehicle in accordance with an example embodiment, and FIGS. **9A-9F** illustrations showing instructional images displayed on a human readable display device for assisting coupling a loader with a work vehicle such a tractor. With reference now to those Figures, a loader latch assist start or commence command is received at **810** by the attach assist apparatus **100**. The attach assist method **800** is initiated at **810** in response to receiving the loader latch assist start or commence command. It is to be appreciated that the loader **13** and the associated work vehicle **1** are initially mutually spaced apart prior to the attach assist apparatus **100** receiving the start or commence command is received at **810**.

(85) A signal may be generated at **820** for instructing the operator to nominally pre-position the tractor relative to the loader in a gross or rough sense, including coupling the portions **50a**, **50b** of the multi-coupler **50** with each other. In an example embodiment, the signal may be generated at **820** for instructing the operator to physically nominally pre-position the tractor relative to the loader may be in the form of one or more instruction images **910**, **920**, **930**, and **940** generated for display on the screen **121** of a human readable display unit **120** and such as shown in FIGS. **9A-9D**.

(86) The method **800** includes displaying one or more orientation image(s) **910**, **920**, **930**, and **940** on the screen **121** of the human readable display unit **120**, wherein the one or more orientation image(s) **910**, **920**, **930**, and **940** are representative of desired orientations of one or more components of the loader informing the operator of the associated vehicle how to move the one or more components of the loader to predetermined positions for pre-positioning the loader in the predetermined orientation to facilitate the coupling. The one or more orientation image(s) **910**, **920**, **930**, and **940** displayed on the screen **121** may include for example information related to one or more of an orientation of a carrier arm **21** of the loader **13**, an orientation of a mast arm **19** of the loader **13**, and/or an orientation of a mounting mast **18** of the loader **13** that is operably coupleable with a corresponding mounting frame **16** affixed to the associated work vehicle **1**.

(87) For example, the image **910** shown in FIG. **9A** displayed on the screen **121** of the human readable display unit **120** includes information relating to a desired spacing and desired vertical orientation between the mounting mast **18** of the loader **13** and the mounting frame **16** on the tractor **1** for ensuring that the orientation of the mounting mast **18** is suitable for easy connection with the mounting frame **16** on the tractor **1**. In this case, one or more electronic service control valves (SCVs) on the loader may be selectively actuated for further pre-positioning the mounting mast portion of the loader in the predetermined orientation to facilitate the coupling. In an example embodiment, a further image **920** (FIG. **9B**) provides instructions to the operator on coupling the first and second portions **50a**, **50b** of the multi-coupler device **50**. It is to be appreciated that the motion of the hydraulic lifting cylinders **24** and/or hydraulic pivoting cylinders **28** may be controlled in accordance with an aspect of the example embodiment by the attach assist apparatus **100** automatically and/or semi-automatically operating the multi-coupler device **50** in accordance with an embodiment while the front loader **13** is detached from the tractor **10** and while the first and second portions **50a**, **50b** may be selectively coupled. This may be used to assist the coupling of the loader with the tractor by ensuring that the orientation of the mounting mast **18** is suitable for easy connection with the mounting frame **16** on the tractor **1** using the hydraulics on the loader to move the loader into the desired orientation and height following coupling the first and second

portions **50a**, **50b** of the multi-coupler device **50**.

(88) As a further example, the image **930** shown in FIG. 9C displayed on the screen **121** of the human readable display unit **120** includes information relating to a desired vertical orientation between the mounting mast **18** of the loader **13** and the mounting frame **16** on the tractor **1** for ensuring that the orientation of the mounting mast **18** is suitable for easy connection with the mounting frame **16** on the tractor **1**. The image instructs the operator that the hydraulic pivoting cylinder **28** may be operated so that the implement or tool **22** carried on the loader and moved thereby in combination with the parking support members **41** of the parking stand unit **40** of the example embodiment can be used to suitably pivot the loader for establishing the desired vertical height of the mounting mast **18** of the loader **13** relative to the mounting frame **16** on the tractor **1**.

(89) As yet a further example, the image **940** shown in FIG. 9D displayed on the screen **121** of the human readable display unit **120** includes information relating to a desired angular orientation between the mounting mast **18** of the loader **13** and the mounting frame **16** on the tractor **1** for ensuring that the orientation of the mounting mast **18** is suitable for easy connection with the mounting frame **16** on the tractor **1**. The image instructs the operator that the pair of hydraulic lifting cylinders **24** may be operated so that the mounting mast **18** of the loader **13** may be moved relative to the mounting frame **16** on the tractor **1** for establishing the desired angular orientation between the mounting mast **18** of the loader **13** and the mounting frame **16** on the tractor **1**.

(90) A signal is received at **830** confirming that the tractor is nominally positioned relative to the loader and that the portions **50 a**, **50 b** of the multi-coupler **50** are mutually coupled with each other. The signal may be received for example from the operator via the touchscreen **134** by the operator using a pointer or finger touching onto a selected portion of the screen **122** while the mounting mast **18** of the loader **13** and the mounting frame **16** of the tractor **1** are in their respective desired positions.

(91) A forward movement instruction signal is generated at **840** for initiating movement of the associated work vehicle **1** towards the loader **13**. In an example, the generating the forward movement instruction signal comprises generating a forward movement instruction image **950** (FIG. 9E) on the screen **121** of a human readable display unit **120**, wherein the forward movement instruction image **950** instructs the operator of the associated work vehicle **1** to initiate the movement of the associated work vehicle **1** towards the loader **13**.

(92) In addition, at **840**, a float condition instruction signal may be generated to establish a float condition in hydraulic lifting cylinders **24** of the loader **13** to relax a mounting mast **18** of the loader **13** that is operably coupleable with a corresponding mounting frame **16** affixed to the associated work vehicle **1** permitting the mounting mast **18** to move relative to the mounting frame **16** during the coupling.

(93) In an example embodiment, the generating the float condition instruction signal at **840** comprises generating a float condition cycle instruction image **950** on the screen **121** of the human readable display unit **120**, wherein the float condition cycle instruction image instructs the operator of the associated work vehicle to intermittently establish, with the loader **13** and the associated work vehicle **1** being in mutual contact and with the associated work vehicle **1** moving towards the loader **13**, the float condition between periods of a locked condition of the mounting mast **18** of the loader **13** wherein in the locked condition movement between the mounting mast **18** and the mounting frame **16** is prevented.

(94) In a further example embodiment, the generating the float condition instruction signal at **840** comprises delivering the float condition cycle instruction signal via a communication interface **119** to a controller the associated work vehicle to automatically intermittently establish, with the loader **13** and the associated work vehicle **1** being in mutual contact and with the associated work vehicle **1** moving towards the loader **13**, the float condition between periods of a locked condition of the mounting mast **18** of the loader **13** wherein in the locked condition movement between the mounting mast **18** and the mounting frame **16** is prevented. The float condition cycle instruction

signal is delivered via the communication interface **119** to a controller **135** the associated work vehicle to automatically intermittently establish, with the loader **13** and the associated work vehicle **1** being in mutual contact and with the associated work vehicle **1** moving towards the loader **13**, the float condition between periods of a locked condition of the mounting mast **18** of the loader **13** wherein in the locked condition movement between the mounting mast **18** and the mounting frame **16** is prevented.

(95) A further confirmation signal may be received at **850** confirming that the mounting mast **18** is suitably and appropriately coupled with the mounting frame **16** on the tractor. This may be based for example on data stored in the non-transient memory device **112** representative of stored values of the signals **71-74** received by the control unit **102** during training of the attach assist apparatus **100**. In an example, the further confirmation signal may comprise the mounting mast inclination signal **73** generated by the mounting mast inclination sensor **63**. In an example, the further confirmation signal may comprise the loader locked signal **75** generated by the loader locked sensor **65**.

(96) A forward movement pause instruction signal is generated at **860** responsive to the attach assist apparatus **100** receiving the coupled confirmation signal and/or in response to a timeout condition that is in response to the forward movement continuing without receiving the confirmation signal beyond a timeout value stored in the non-transient memory device **112**. In an example, the forward movement pause instruction signal comprises one or more forward movement pause instruction images **960** (FIG. **9F**) generated on the screen **121** of the human readable display unit **120**, wherein the forward movement pause instruction images **960** instruct the operator of the associated work vehicle **1** to pause the movement of the associated work vehicle **1** towards the loader **13**.

(97) In a further example, the generating the forward movement pause instruction signal at **860** comprises delivering, responsive to receiving the coupled confirmation signal, the forward movement pause instruction signal via the communication interface **119** to the controller the associated work vehicle to automatically pause the movement of the associated work vehicle **1** towards the loader **13**.

(98) Overall and in accordance with an example embodiment, the generating the pre-position instruction signal comprises delivering the pre-position instruction signal via a communication interface **119** to a controller the associated work vehicle to automatically pre-position the loader in the predetermined orientation to facilitate the coupling. In addition, the generating the forward movement instruction signal comprises delivering the pre-position instruction signal via the communication interface **119** to the controller the associated work vehicle to automatically initiate the movement of the associated work vehicle **1** towards the loader **13**. In further addition, the receiving the coupled confirmation signal comprises automatically receiving the coupled confirmation signal as a loader locked signal **75** from a loader locked sensor **65** disposed at an interface between a mounting mast **18** of the loader **13** that is operably coupleable with a corresponding mounting frame **16** affixed to the associated work vehicle **1**, the loader locked signal **75** from the loader locked sensor **65** being representative of a confirmation of the loader **13** being coupled with the associated work vehicle **1**. In still further addition, the generating the forward movement pause instruction signal comprises delivering, responsive to receiving the coupled confirmation signal, the forward movement pause instruction signal via the communication interface **119** to the controller the associated work vehicle to automatically pause the movement of the associated work vehicle **1** towards the loader **13**.

(99) A decision is made at **870** on whether the sequence from **840-860** should be repeated such as for example, if the mounting mast **18** does not fully and completely attach with the mounting frame **16**.

(100) It is to be understood that other embodiments will be utilized and structural and functional changes will be made without departing from the scope of the present invention. The foregoing

descriptions of embodiments of the present invention have been presented for the purposes of illustration and description. It is not intended to be exhaustive or to limit the invention to the precise forms disclosed. Accordingly, many modifications and variations are possible in light of the above teachings. It is therefore intended that the scope of the invention be limited not by this detailed description.

Claims

1. An attach assist apparatus for assisting coupling a loader with an associated work vehicle, the apparatus comprising: an attach assist control unit comprising: a processor device; a non-transient memory device operatively coupled with the processor device; and attach assist control logic stored in the memory device; a human readable display unit operably coupled with the attach assist control unit, the human readable display unit comprising a screen operable to display images that are viewable by an operator of the associated work vehicle; and an input device operably coupled with the attach assist control unit, wherein the processor device is operable to execute the attach assist control logic to: generate a pre-position instruction image and display the pre-position instruction image on the screen of the human readable display unit, wherein the pre-position instruction image informs an operator of the associated vehicle how to pre-position the loader in a predetermined orientation to facilitate the coupling; generate a forward movement instruction image and display the forward movement instruction image on the screen of the human readable display unit, wherein the forward movement instruction image instructs the operator of the associated work vehicle to initiate movement of the associated work vehicle towards the loader; receive a coupled confirmation signal by the input device from the operator of the associated vehicle, the coupled confirmation signal confirming by the operator that the loader is coupled with the associated work vehicle; and generate, responsive to receiving the coupled confirmation signal, a forward movement pause instruction image and display the forward movement pause instruction image on the screen of the human readable display unit, wherein the forward movement pause instruction image instructs the operator of the associated work vehicle to pause the movement of the associated work vehicle towards the loader.
2. The attach assist apparatus according to claim 1, wherein: the processor device is operable to execute the attach assist control logic to: display an orientation image on the screen of the human readable display unit, wherein the orientation image is representative of desired orientations of one or more components of the loader informing the operator of the associated vehicle how to move the one or more components of the loader to predetermined positions for pre-positioning the loader in the predetermined orientation to facilitate the coupling.
3. The attach assist apparatus according to claim 2, wherein: the processor device is operable to execute the attach assist control logic to: display the orientation image on the screen of the human readable display unit as one or more of: an orientation of a carrier arm of the loader; an orientation of a mast arm of the loader; and/or an orientation of a mounting mast of the loader that is operably coupleable with a corresponding mounting frame affixed to the associated work vehicle.
4. The attach assist apparatus according to claim 1, further comprising: a sensor input unit operably coupled with the attach assist control unit; and a loader locked sensor operably coupled with the attach assist control unit by the sensor input unit, the loader locked sensor being disposed at an interface between a mounting mast of the loader that is operably coupleable with a corresponding mounting frame affixed to the associated work vehicle, wherein the processor device is operable to execute the attach assist control logic to receive the coupled confirmation signal by receiving a loader locked signal from the loader locked sensor as an automated coupled confirmation signal representative of the confirmation of the loader being coupled with the associated work vehicle, wherein the processor device is operable to execute the attach assist control logic to generate the forward movement pause instruction for pausing the movement of the associated work vehicle

towards the loader based on receiving the automated coupled confirmation signal.

5. The attach assist apparatus according to claim 1, wherein the processor device is operable to execute the attach assist control logic to: generate a float condition instruction signal to establish a float condition in hydraulic lifting cylinders of the loader to relax a mounting mast of the loader that is operably coupleable with a corresponding mounting frame affixed to the associated work vehicle permitting the mounting mast to move relative to the mounting frame during the coupling.

6. The attach assist apparatus according to claim 5, wherein: the processor device is operable to execute the attach assist control logic to generate the float condition instruction signal by: generating a float condition cycle instruction image as the float condition instruction signal and displaying the float condition cycle instruction image on the screen of the human readable display unit, wherein the float condition cycle instruction image instructs the operator of the associated work vehicle to intermittently establish, with the loader and the associated work vehicle being in mutual contact and with the associated work vehicle moving towards the loader, the float condition between periods of a locked condition of the mounting mast of the loader wherein in the locked condition movement between the mounting mast and the mounting frame is prevented.

7. The attach assist apparatus according to claim 5, further comprising: a control signal output unit operably coupled with the attach assist control unit; and a valve operably coupled with the attach assist control unit by the control signal output unit, the valve being responsive to a control signal generated by the attach assist apparatus to actuate to selectively relieve hydraulic fluid from a chamber of one or more hydraulic lifting cylinders of the loader, wherein the processor device is operable to execute the attach assist control logic to: generate the float condition instruction signal as the control signal to establish a float condition in hydraulic lifting cylinders of the loader to relax a mounting mast of the loader that is operably coupleable with a corresponding mounting frame affixed to the associated work vehicle permitting the mounting mast to move relative to the mounting frame during the coupling.

8. An attach assist apparatus for assisting coupling a loader with an associated work vehicle, the apparatus comprising: an attach assist control unit comprising: a processor device; a non-transitory memory device operatively coupled with the processor device; and attach assist control logic stored in the non-transitory memory device; a communication interface operably coupling the attach assist control unit with a controller of the associated work vehicle; and a loader locked sensor operably coupled with the attach assist control unit, the loader locked sensor being disposed at an interface between a mounting mast of the loader that is operably coupleable with a corresponding mounting frame affixed to the associated work vehicle, wherein the processor device is operable to execute the attach assist control logic to: generate a pre-position instruction signal for automatically pre-positioning the loader in a predetermined orientation to facilitate the coupling of the loader with the associated work vehicle; deliver the instruction signal via the communication interface to the controller the associated work vehicle to automatically pre-position the loader in the predetermined orientation to facilitate the coupling of the loader with the associated work vehicle; generate a forward movement instruction signal for automatically initiating movement of the associated work vehicle towards the loader; deliver the forward movement instruction signal via the communication interface to the controller of the associated work vehicle to automatically initiate the movement of the associated work vehicle towards the loader; automatically receive a loader locked signal from the loader locked sensor, the loader locked signal being representative of a confirmation of the loader being coupled with the associated work vehicle; and deliver, responsive to receiving the loader locked signal, a forward movement pause instruction signal via the communication interface to the controller of the associated work vehicle to automatically pause the movement of the associated work vehicle towards the loader.

9. The attach assist apparatus according to claim 8, wherein: the processor device is operable to execute the attach assist control logic to: generate a float condition instruction signal to automatically establish a float condition in hydraulic lifting cylinders of the loader to relax a

mounting mast of the loader that is operably coupleable with a corresponding mounting frame affixed to the associated work vehicle; and deliver the float condition instruction signal via the communication interface to the controller of the associated work vehicle to automatically establish the float condition in the hydraulic lifting cylinders of the loader to relax the mounting mast of the loader thereby permitting the mounting mast to move relative to the mounting frame during the coupling.

10. The attach assist apparatus according to claim 8, further comprising: a control signal output unit operably coupled with the attach assist control unit; and a valve operably coupled with the attach assist control unit by the control signal output unit, the valve being responsive to a control signal generated by the attach assist apparatus to establish a float condition by selectively relieving hydraulic fluid from a chamber of one or more hydraulic lifting cylinders of the loader, wherein the processor device is operable to execute the attach assist control logic to: generate the control signal to establish the float condition in hydraulic lifting cylinders of the loader to relax a mounting mast of the loader that is operably coupleable with a corresponding mounting frame affixed to the associated work vehicle permitting the mounting mast to move relative to the mounting frame during the coupling.

11. A method for assisting coupling a loader with an associated work vehicle, the method comprising: with the loader and the associated work vehicle mutually spaced apart: generating a pre-position instruction image on a screen of a human readable display unit, wherein the pre-position instruction image informs an operator of the associated vehicle how to pre-position the loader in a predetermined orientation to facilitate the coupling; and generating a forward movement instruction image on the screen of a human readable display unit, wherein the forward movement instruction image instructs the operator of the associated work vehicle to initiate movement of the associated work vehicle towards the loader; and with the loader and the associated work vehicle being in mutual contact: receiving a coupled confirmation signal from the operator of the associated vehicle via an input device confirming by the operator that the loader is coupled with the associated work vehicle; and responsive to receiving the coupled confirmation signal, generating a forward movement pause instruction image on the screen of the human readable display unit, wherein the forward movement pause instruction image instructs the operator of the associated work vehicle to pause the movement of the associated work vehicle towards the loader.

12. The method according to claim 11, further comprising: displaying an orientation image on the screen of the human readable display unit, wherein the orientation image is representative of desired orientations of one or more components of the loader informing the operator of the associated vehicle how to move the one or more components of the loader to predetermined positions for pre-positioning the loader in the predetermined orientation to facilitate the coupling.

13. The method according to claim 12, wherein the displaying the orientation image comprises: displaying an orientation image on the screen of the human readable display unit of one or more of: an orientation of a carrier arm of the loader; an orientation of a mast arm of the loader; and/or an orientation of a mounting mast of the loader that is operably coupleable with a corresponding mounting frame affixed to the associated work vehicle.

14. The method according to claim 11, wherein: the receiving the coupled confirmation signal comprises: receiving the coupled confirmation signal as an automated loader locked signal from a loader locked sensor disposed at an interface between a mounting mast of the loader that is operably coupleable with a corresponding mounting frame affixed to the associated work vehicle, the automated loader locked signal from the loader locked sensor being representative of an automated confirmation of the loader being coupled with the associated work vehicle; and the generating the forward movement pause instruction image comprises: responsive to receiving the automated coupled confirmation signal, generating the forward movement pause instruction image on the screen of the human readable display unit for instructing the operator of the associated work vehicle to pause the movement of the associated work vehicle towards the loader.

15. The method according to claim 11, further comprising: with the loader and the associated work vehicle being in mutual contact and with the associated work vehicle moving towards the loader: generating a float condition instruction signal to establish a float condition in hydraulic lifting cylinders of the loader to relax a mounting mast of the loader that is operably coupleable with a corresponding mounting frame affixed to the associated work vehicle permitting the mounting mast to move relative to the mounting frame during the coupling.

16. The method according to claim 15, wherein: the generating the float condition instruction signal comprises: generating a float condition cycle instruction image on the screen of the human readable display unit, wherein the float condition cycle instruction image instructs the operator of the associated work vehicle to intermittently establish, with the loader and the associated work vehicle being in mutual contact and with the associated work vehicle moving towards the loader, the float condition between periods of a locked condition of the mounting mast of the loader wherein in the locked condition movement between the mounting mast and the mounting frame is prevented.

17. The method according to claim 15, wherein: the generating the float condition instruction signal comprises: delivering the float condition cycle instruction signal via a communication interface to a controller the associated work vehicle to automatically intermittently establish, with the loader and the associated work vehicle being in mutual contact and with the associated work vehicle moving towards the loader, the float condition between periods of a locked condition of the mounting mast of the loader wherein in the locked condition movement between the mounting mast and the mounting frame is prevented.

18. A method for assisting coupling a loader with an associated work vehicle, the method comprising: with the loader and the associated work vehicle mutually spaced apart: generating a pre-position instruction signal to pre-position the loader in a predetermined orientation to facilitate the coupling by delivering the pre-position instruction signal via a communication interface to a controller the associated work vehicle to automatically pre-position the loader in the predetermined orientation to facilitate the coupling; generating a forward movement instruction signal for initiating movement of the associated work vehicle towards the loader by delivering the pre-position instruction signal via the communication interface to the controller the associated work vehicle to automatically initiate the movement of the associated work vehicle towards the loader; with the loader and the associated work vehicle being in mutual contact: receiving a coupled confirmation signal representative of a confirmation of the loader being coupled with the associated work vehicle by automatically receiving the coupled confirmation signal as a loader locked signal from a loader locked sensor disposed at an interface between a mounting mast of the loader that is operably coupleable with a corresponding mounting frame affixed to the associated work vehicle, the loader locked signal from the loader locked sensor being representative of the confirmation of the loader being coupled with the associated work vehicle; and responsive to receiving the coupled confirmation signal, delivering forward movement pause instruction signal via the communication interface to the controller the associated work vehicle to automatically pause the movement of the associated work vehicle towards the loader.

19. The method according to claim 18, further comprising: with the loader and the associated work vehicle being in mutual contact and with the associated work vehicle moving towards the loader: delivering a float condition instruction signal via the communication interface to the controller the associated work vehicle to automatically establish a float condition in hydraulic lifting cylinders of the loader to relax a mounting mast of the loader that is operably coupleable with a corresponding mounting frame affixed to the associated work vehicle permitting the mounting mast to move relative to the mounting frame during the coupling.

20. The method according to claim 18, further comprising: with the loader and the associated work vehicle being in mutual contact and with the associated work vehicle moving towards the loader: delivering a float condition cycle instruction signal via a communication interface to the controller the associated work vehicle to automatically intermittently establish, with the loader and the

associated work vehicle being in mutual contact and with the associated work vehicle moving towards the loader, the float condition between periods of a locked condition of the mounting mast of the loader wherein in the locked condition movement between the mounting mast and the mounting frame is prevented.
